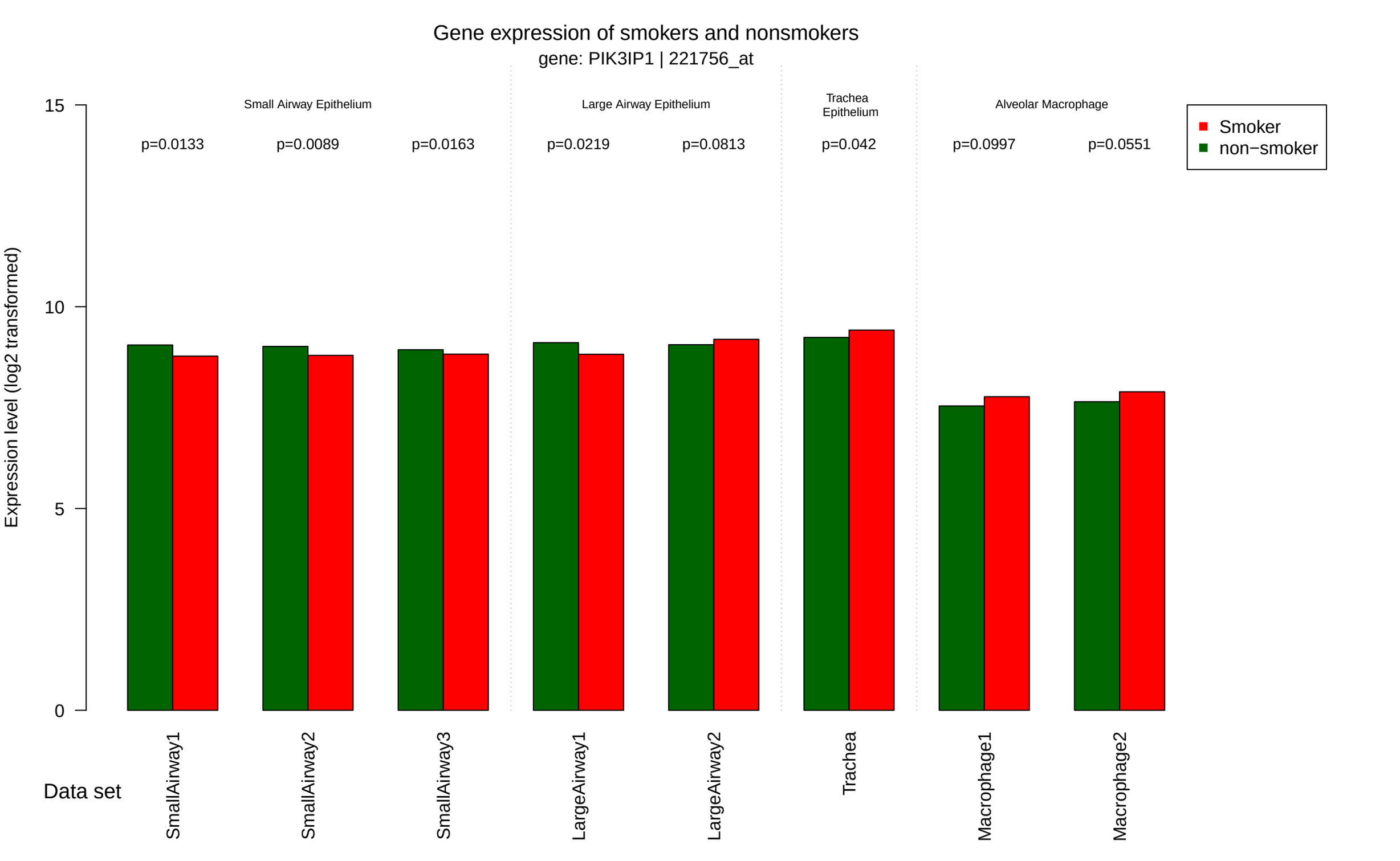
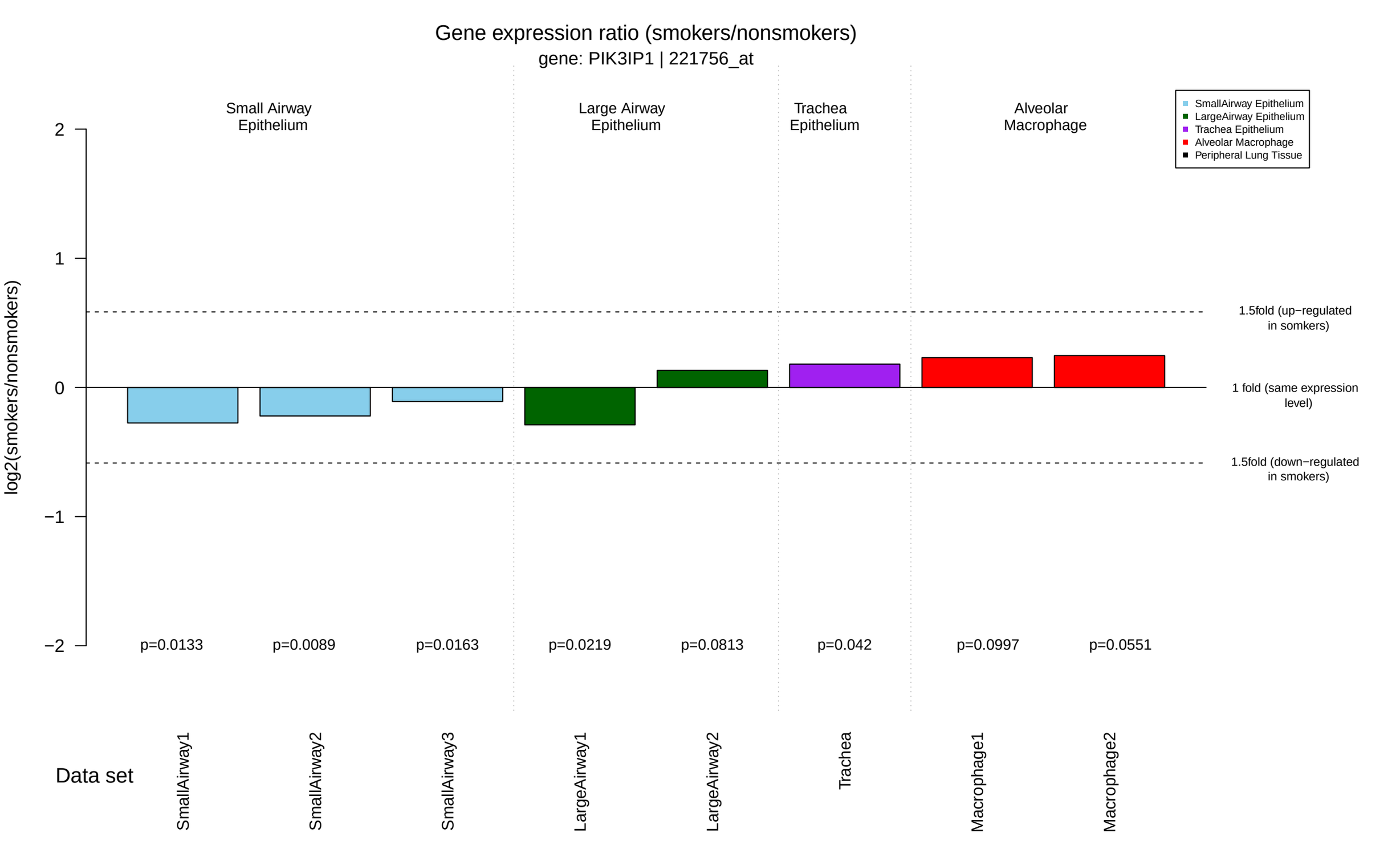
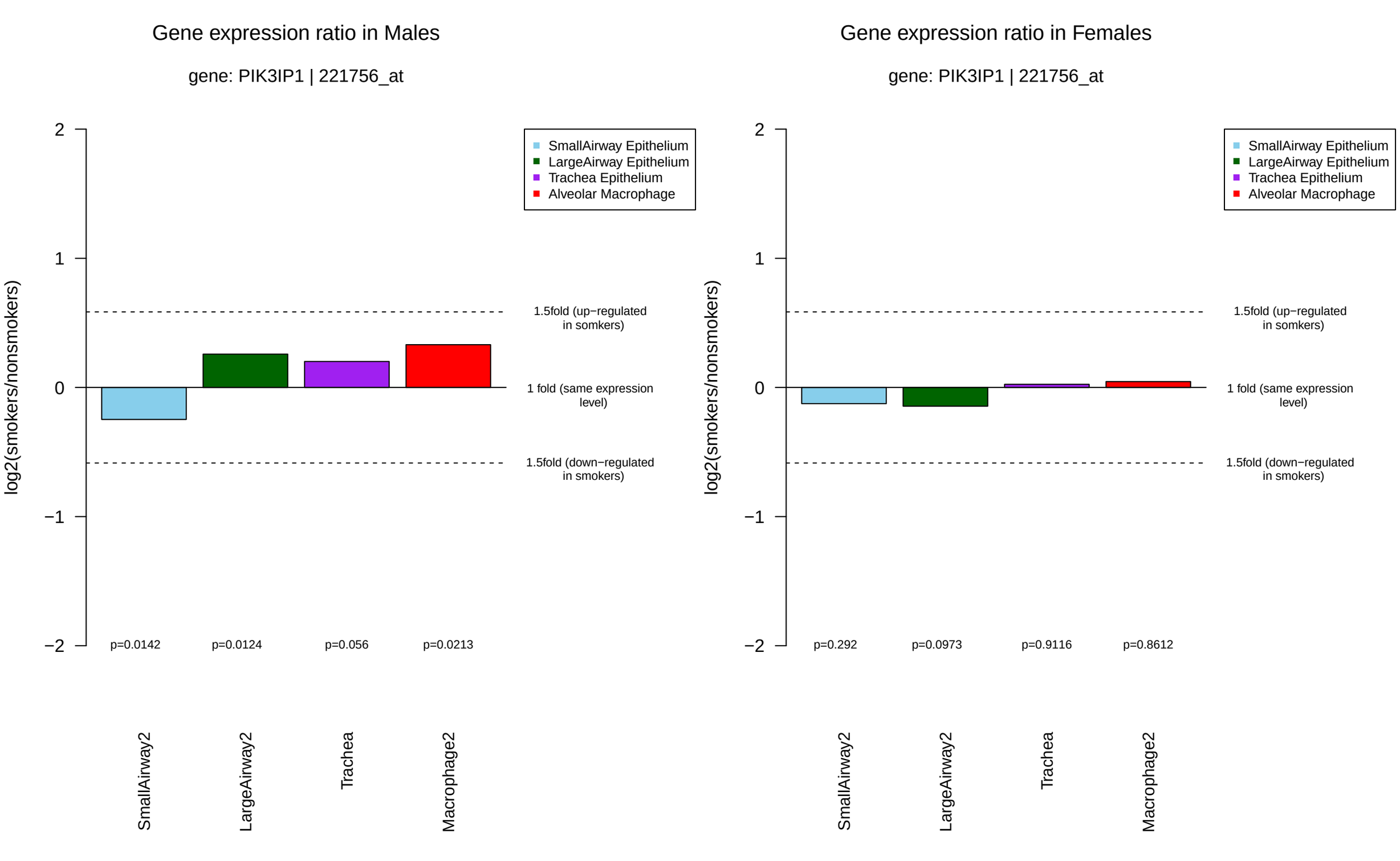
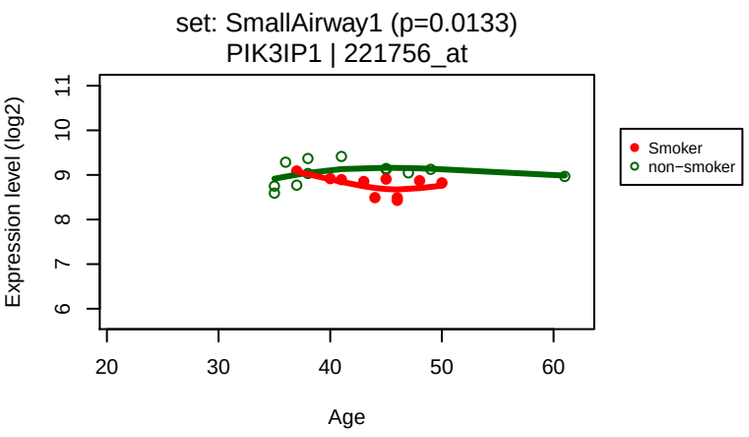
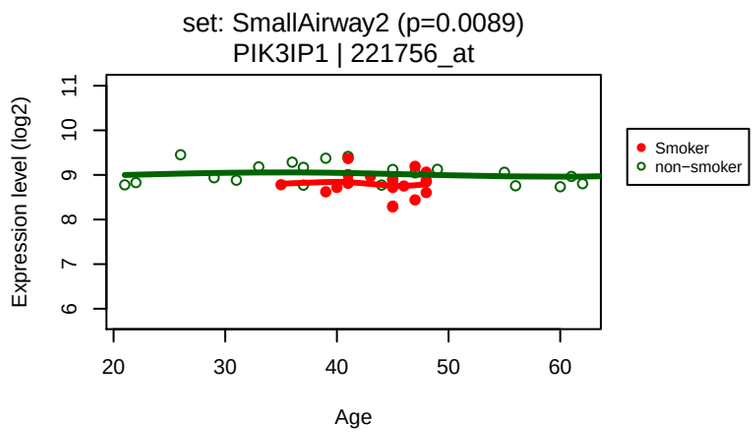
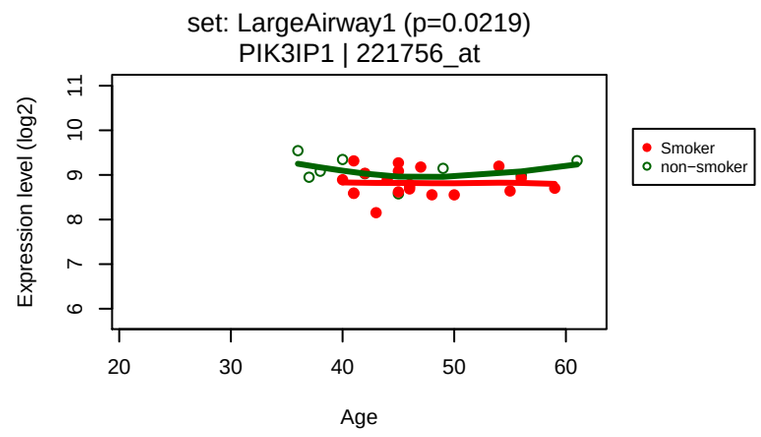

Gene expression of smokers and nonsmokers
gene: PIK3IP1 | 221756_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0133
p=0.0089
p=0.0163
p=0.0219
p=0.0813
p=0.042
p=0.0997
p=0.0551
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

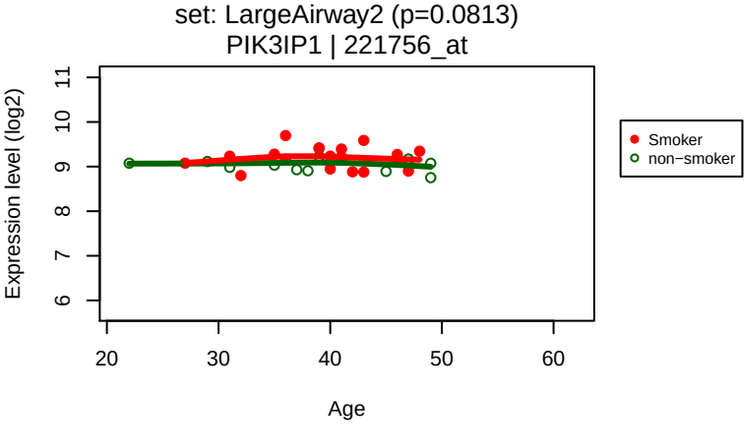

Gene expression ratio (smokers/nonsmokers)
gene: PIK3IP1 | 221756_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0133
p=0.0089
p=0.0163
p=0.0219
p=0.0813
p=0.042
p=0.0997
p=0.0551
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

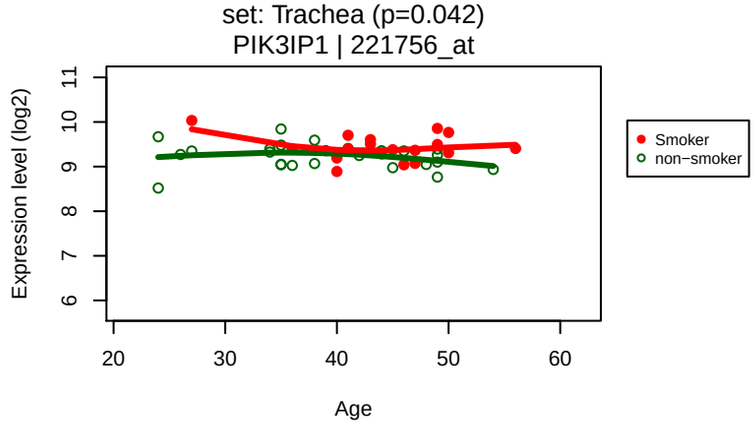

Gene expression ratio in Males
gene: PIK3IP1 | 221756_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0142
p=0.0124
p=0.056
p=0.0213
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PIK3IP1 | 221756_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.292
p=0.0973
p=0.9116
p=0.8612
SmallAirway2
LargeAirway2
Trachea
Macrophage2

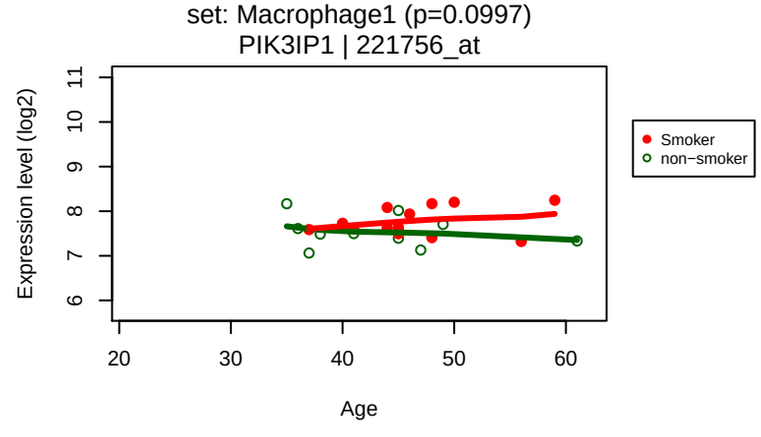

set: SmallAirway1 (p=0.0133)
PIK3IP1 | 221756_at
Expression level (log2)
Age
Smoker
non−smoker

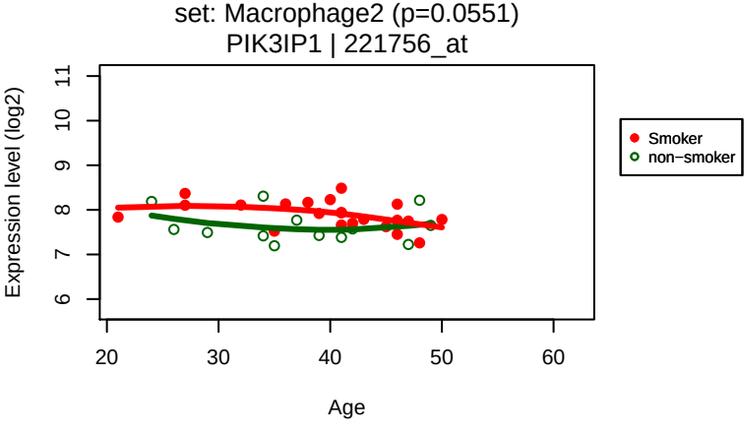

set: SmallAirway2 (p=0.0089)
PIK3IP1 | 221756_at
Expression level (log2)
Age
Smoker
non−smoker

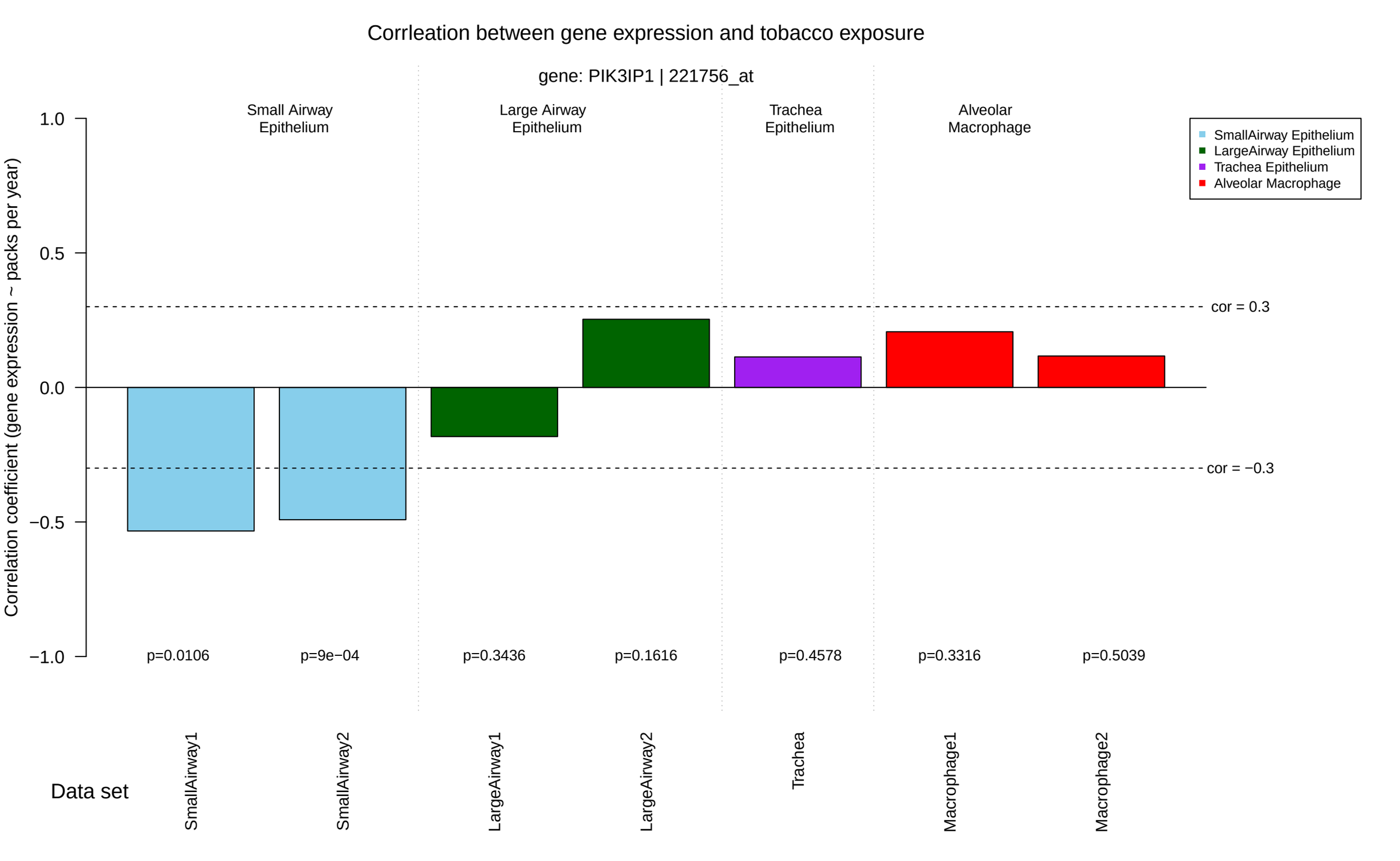

set: LargeAirway1 (p=0.0219)
PIK3IP1 | 221756_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0813)
PIK3IP1 | 221756_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.042)
PIK3IP1 | 221756_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0997)
PIK3IP1 | 221756_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0551)
PIK3IP1 | 221756_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PIK3IP1 | 221756_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0106
p=9e−04
p=0.3436
p=0.1616
p=0.4578
p=0.3316
p=0.5039
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2